

Software design and testing C# SDT621 PRACTICAL MODULE - PM-04 2026

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20240611

20240 SAQA ID NQF 6 611

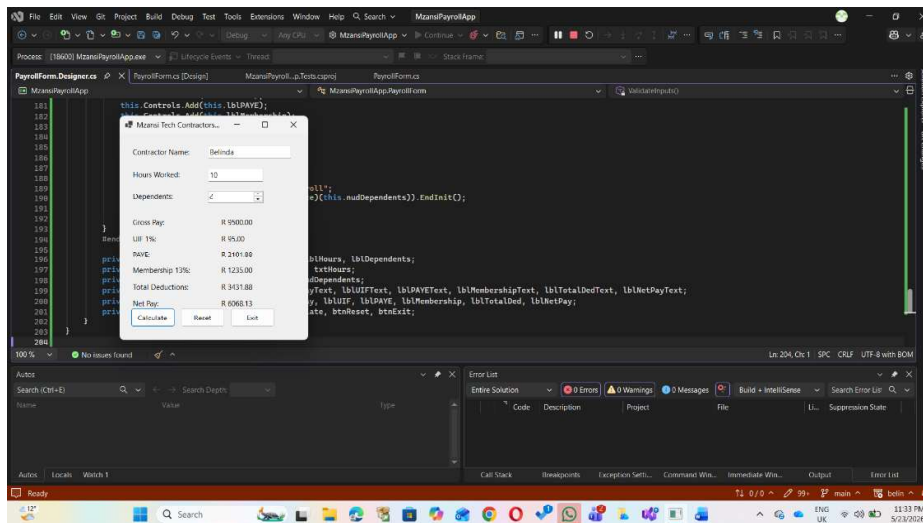
Unit Testing

Test Case	Input	Expected Output	Result
Gross Pay	10	R9500.00	Pass
UIF	R9500.00	R95.00	Pass
Membership	R9500.00	R1235.00	Pass
PAYE	Gross=R9500 Dependents =2	R2114.38	Pass
Net Pay	Gross- all deductions	R6055.62	Pass

Integration Testing

Purpose of integration testing is to ensure that all the inputs, calculations and output sections work together correctly.

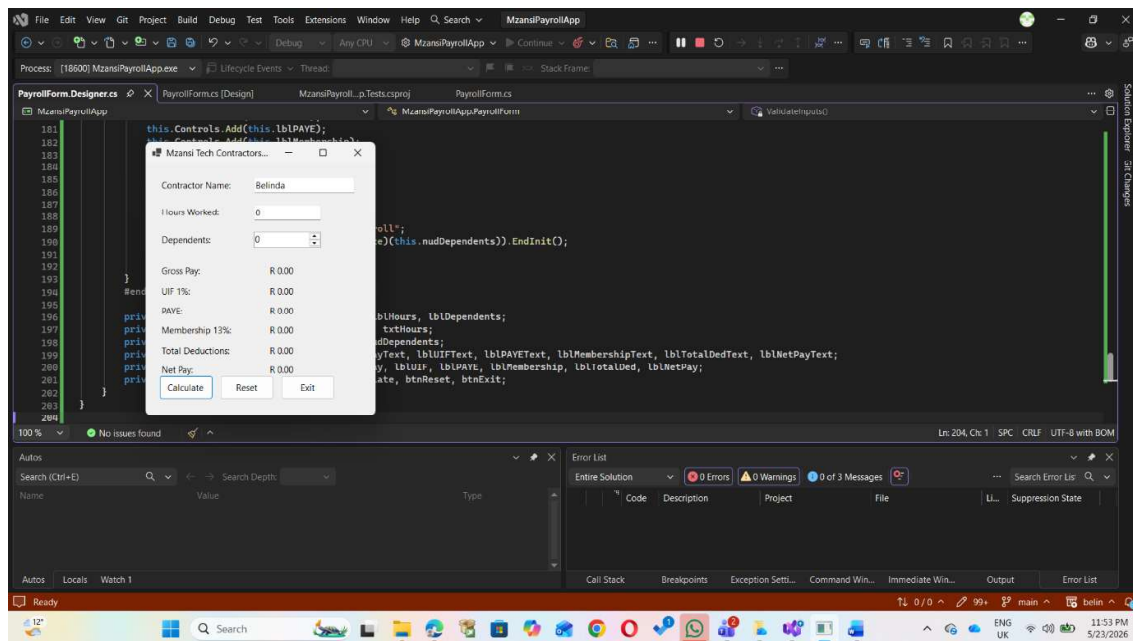
Normal Calculation



Actual Results: All the values displayed correctly in the application.

Results: Pass

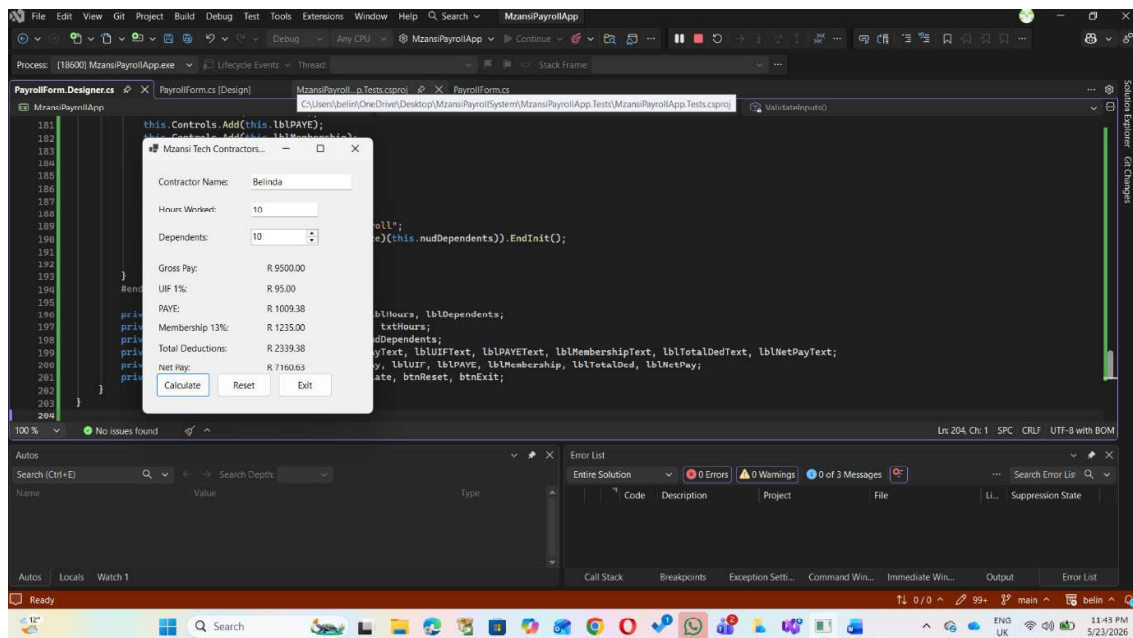
Zero Hours and Dependents



Actual Results: Application displayed R0.00 for all deductions and Net Pay.

Results: Pass

Maximum Dependents



Actual Result: Application capped PAYE at 0 when base went negative and calculated Net Pay correctly.

Results: Pass

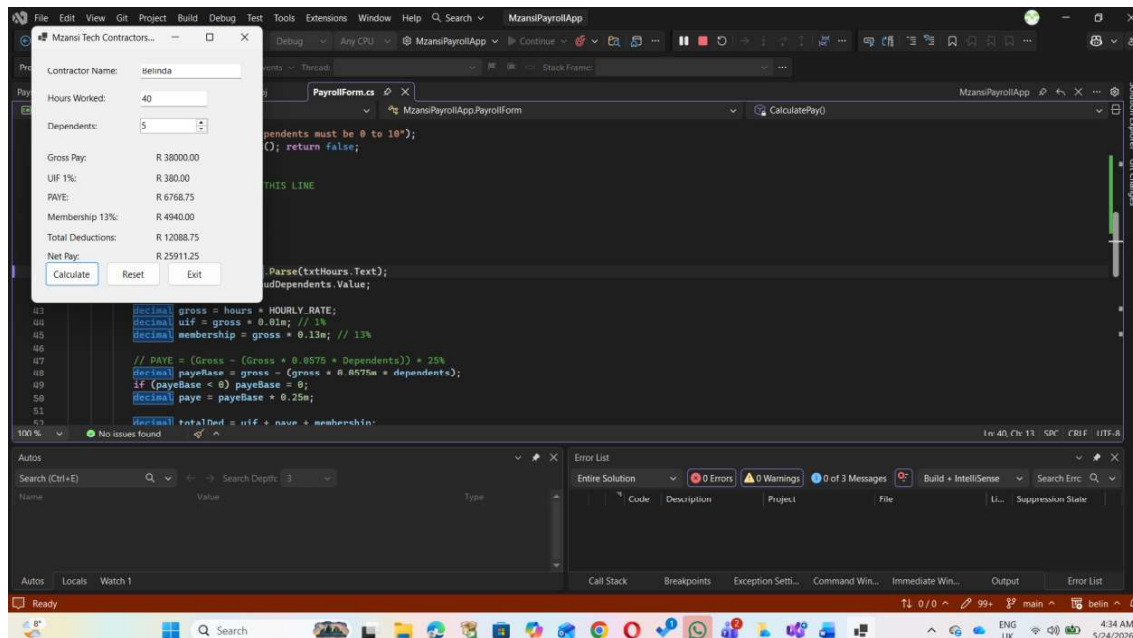
Conclusion:

All integration test passed. The application correctly passes Gross Pay into UIF,PAYE, and Membership calculations, and combines deductions to calculate Net Pay.

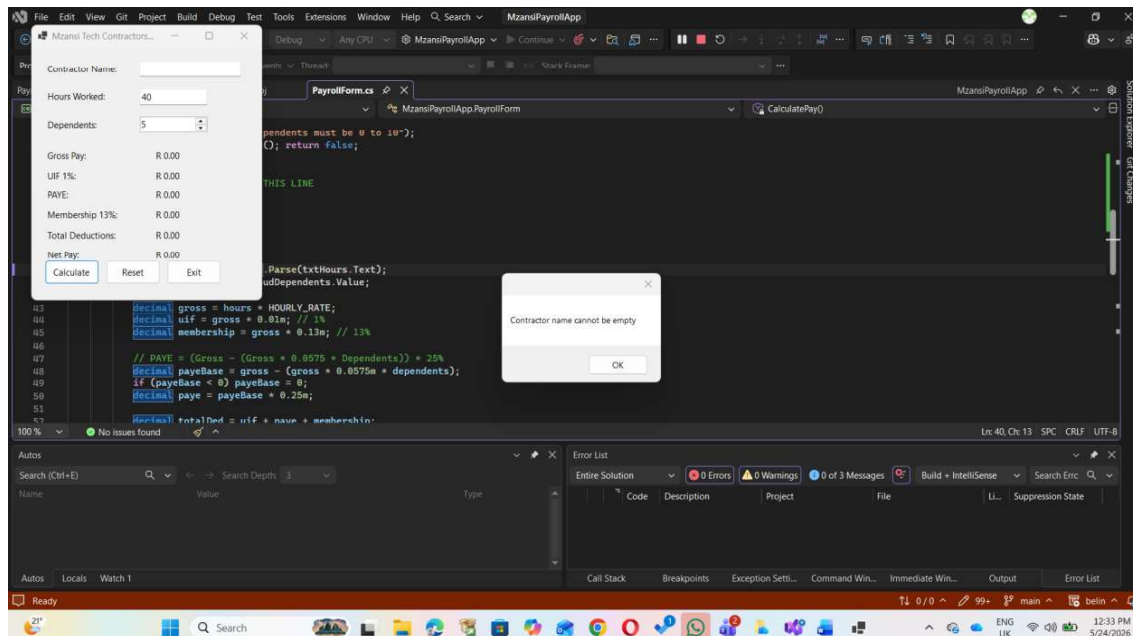
System Testing

Step	Action	Expected Results
1	Enter Name: Belinda, Hours : 40, Dependents: 5. Click Calculate	Gross R38.000, UIF R380.00, PAYE R14,440.00, Membership R0.00, Net R23.180.
2	Enter Name: , Hours : 40, Dependents: 5. Click Calculate	Popup:Name cannot be empty
3	Enter Name: Belinda , Hours: -5, Dependents: 1. Click Calculate	No popup: 15 auto-corrects to 10
4	Enter Name: Belinda , Hours: 15, Dependents: 15. Click Calculate	Popup: Dependents must be 0-10
5	Click Reset	All fields clear, Net/Gross= R0.00

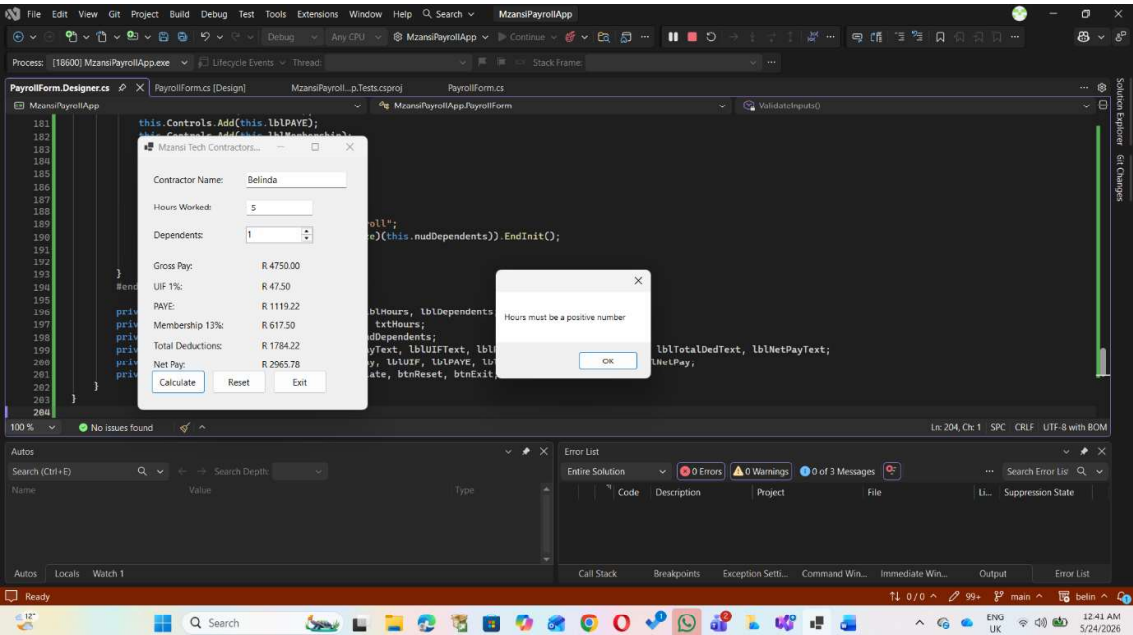
(Step1)



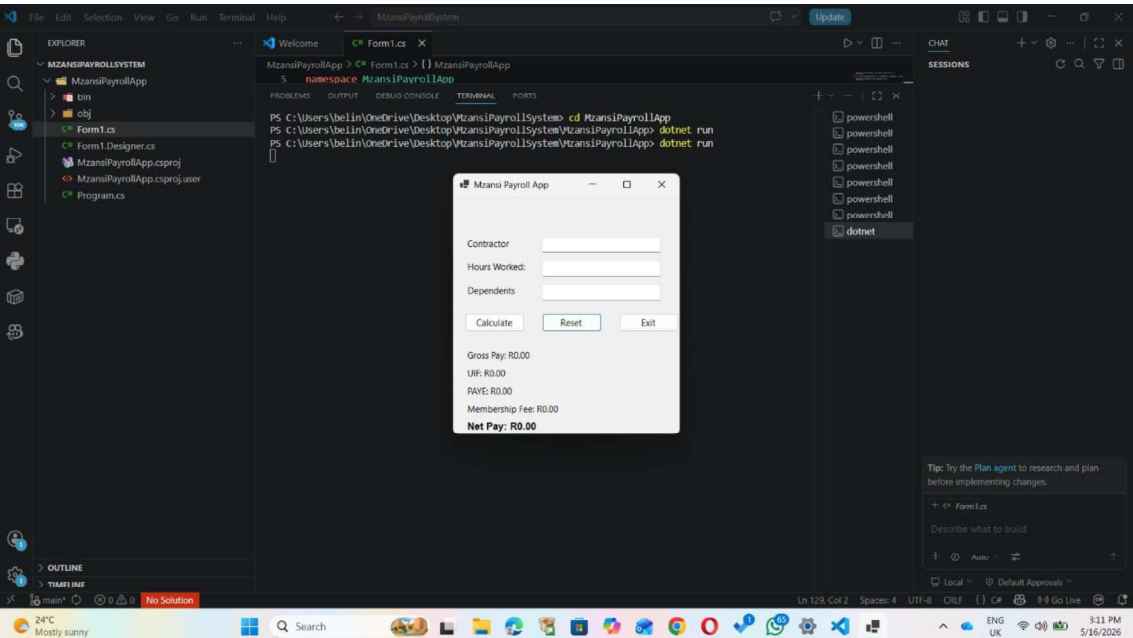
(Step 2)



(Step 3)

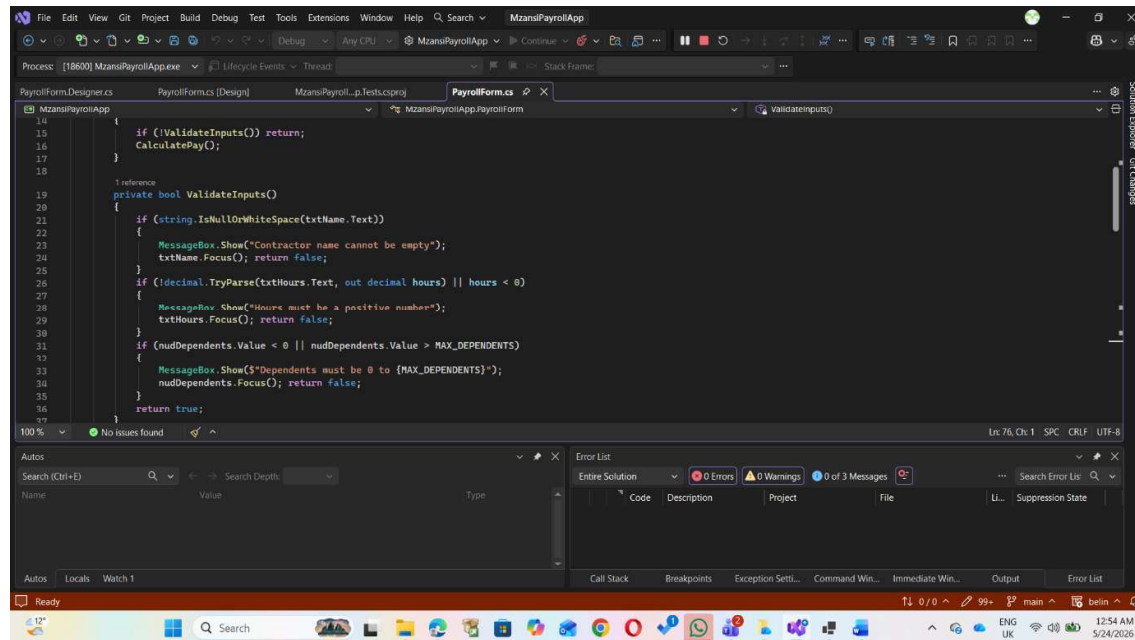


(Step 5)

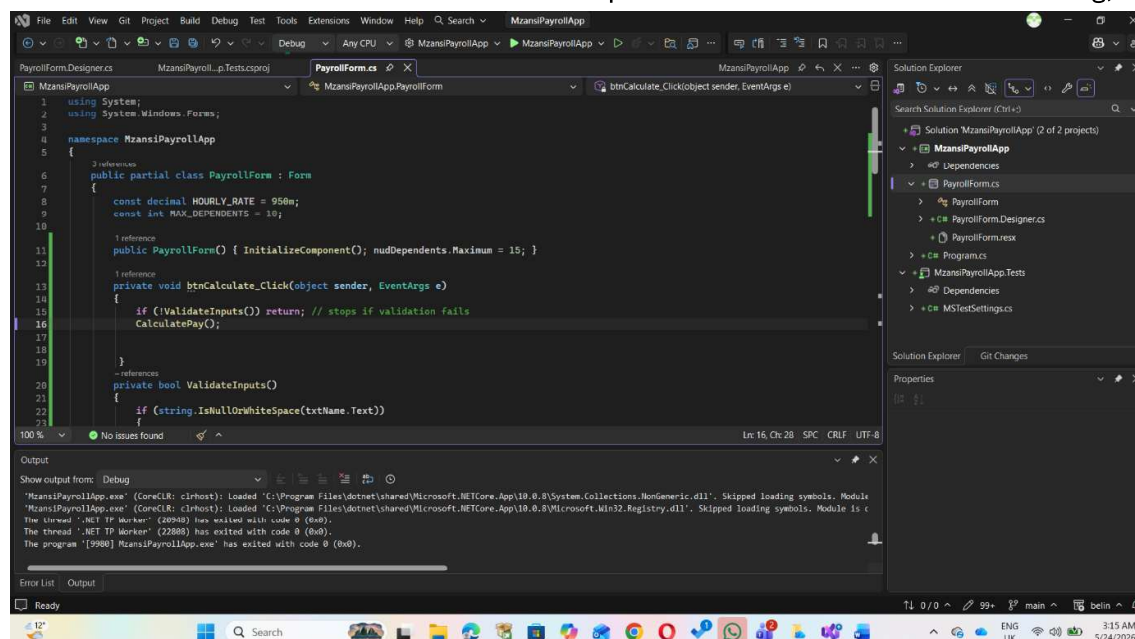


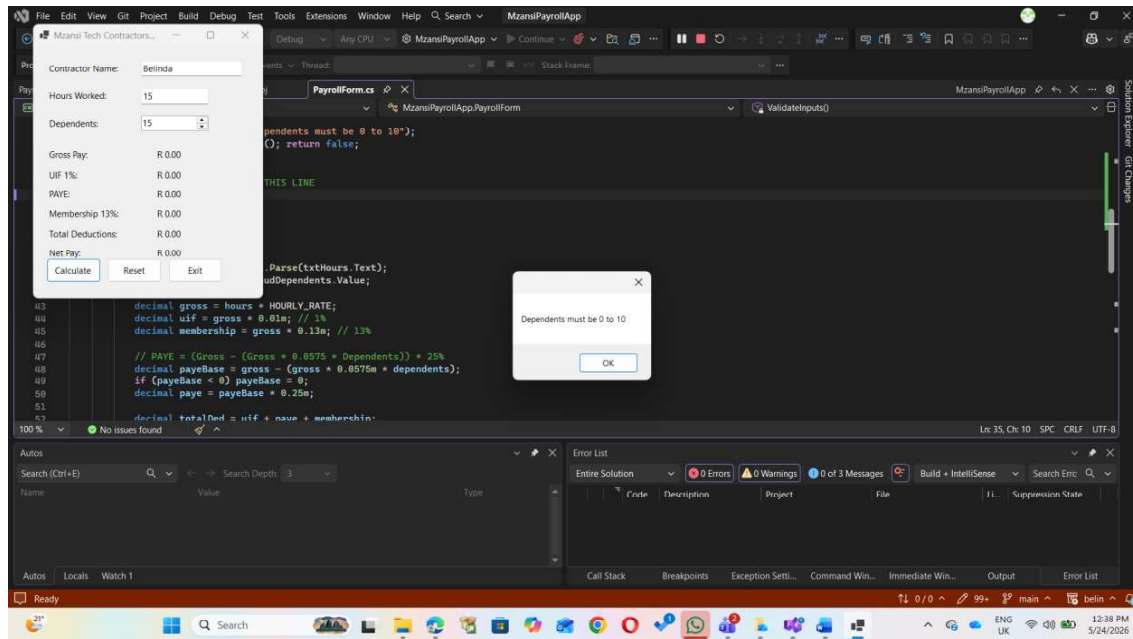
Retesting

Default Code: nudDependents.Maximum = 10. The control auto-corrected 15 to 10, so validation could not be tested.



Corrected Code: nudDependents.Maximum = 15 was set temporarily to allow testing of the validation rule. Validation now checks if dep > 10 and shows an error. After testing,



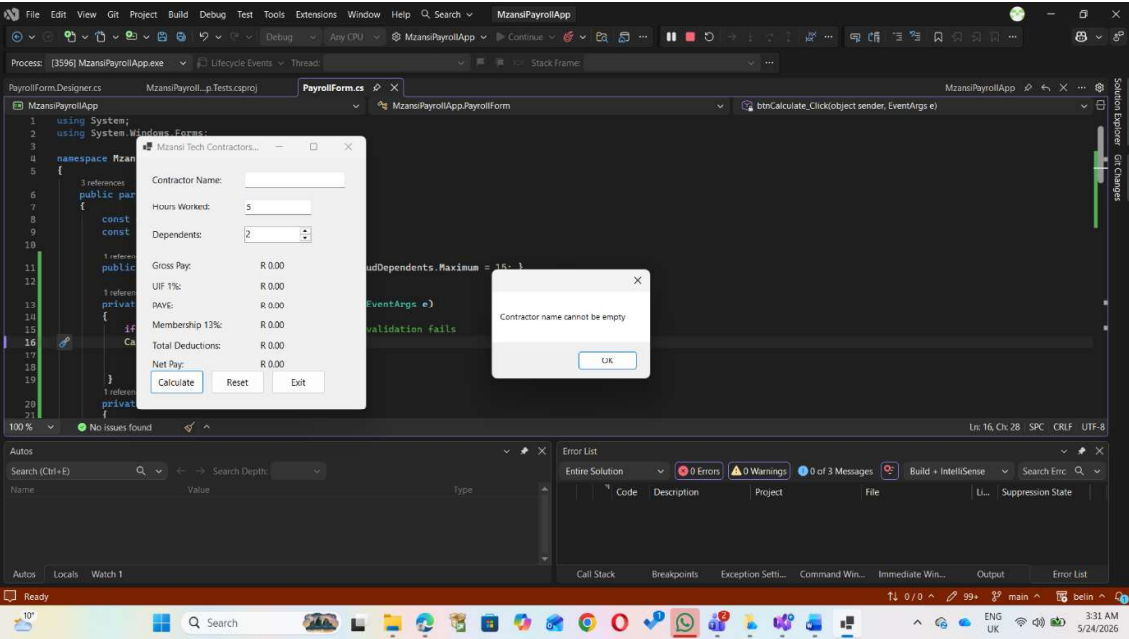


Regression Testing

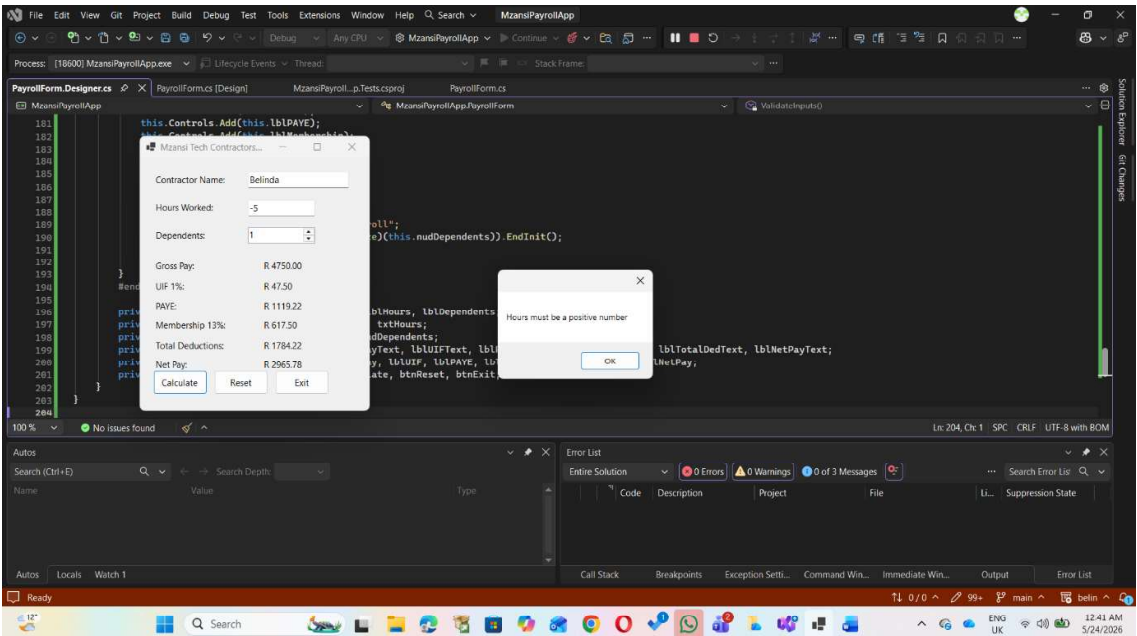
Test ID	Validation Rule	Test Data	Expected Result	Actual Result	Defect Found	Correction Made	Retest	Regression
T1	No empty contract or name	Name =	Error: Name required	Pop up Shown	App allowed blank	Added if string.IsNullOrEmpty	Pass	Pass
T2	No negative hours	Hours = -5	Error: Hours positive	Pop up Shown	App allowed -5	App if hours < 0	Pass	Pass
T3	No non-numeric hours	Hours = abc	Error: Enter number	Pop up Shown	App Crashed	Added TryParse	Pass	Pass
T4	Dependents = 0-10 only	Dependents = 15	Error: Max 10	Pop up Shown	App auto-corrected 15 to 10	Temp set nudDependents. Maximum = 15 + added if dep > 10	Pass	Pass

T5	Reset button works	Click Rest	All fields = 0	All Cleared	None	None	-	Pass
T6	Valid data	Name= Belinda , Hours=40, Dep=5	Calculate Gross / Net	Correct values shown	None	None	-	Pass

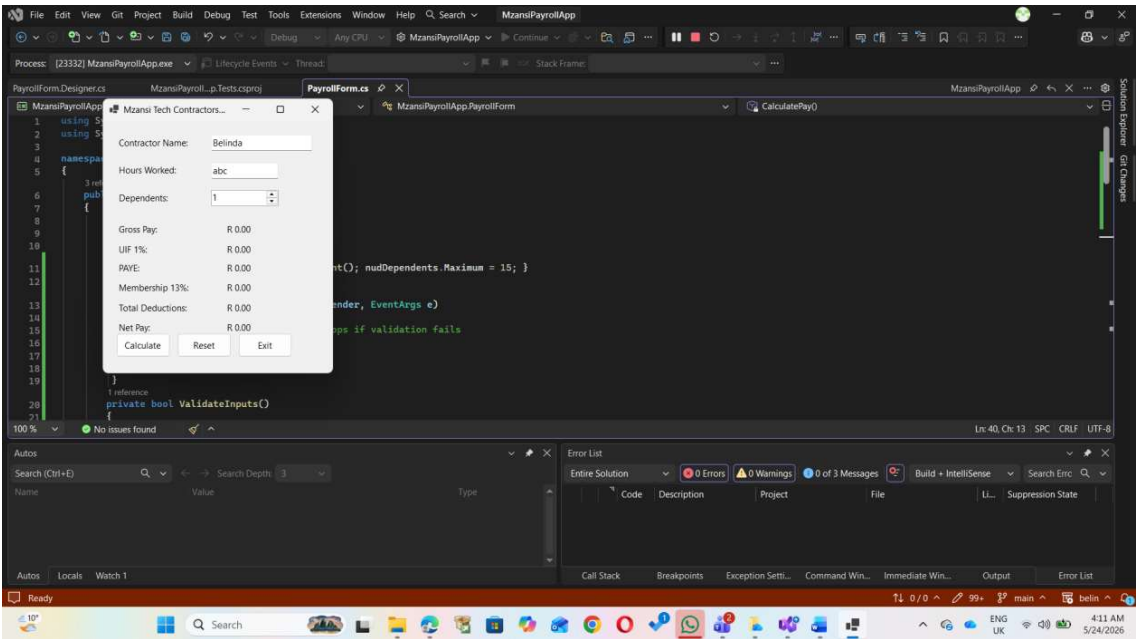
T1-Empty Name

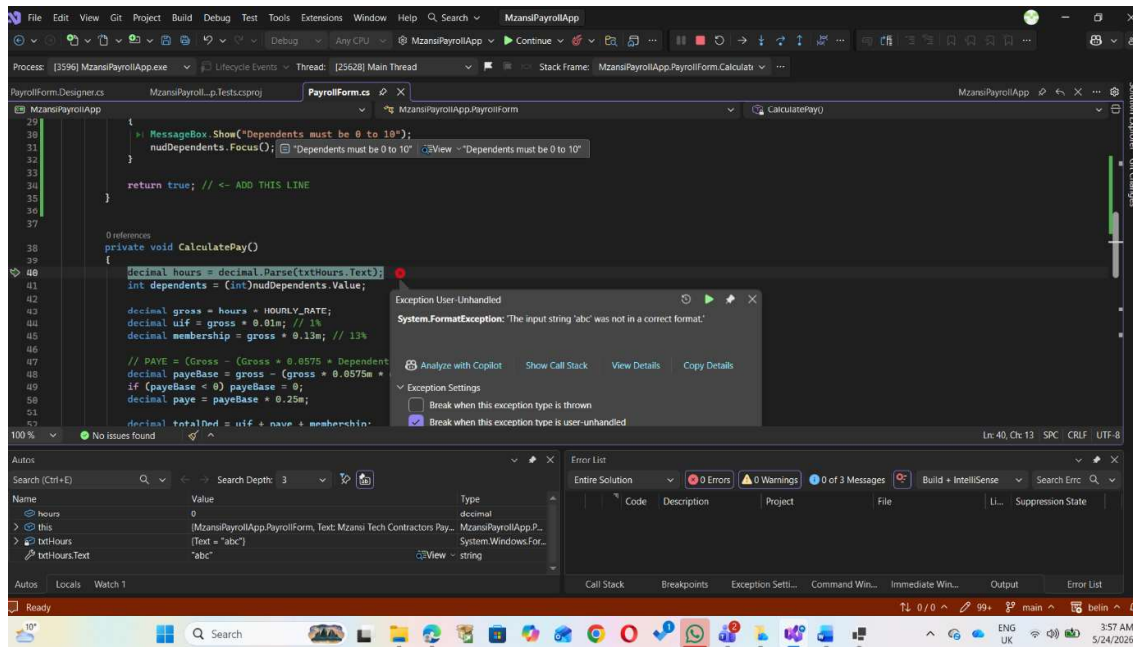


T2- No negative hours

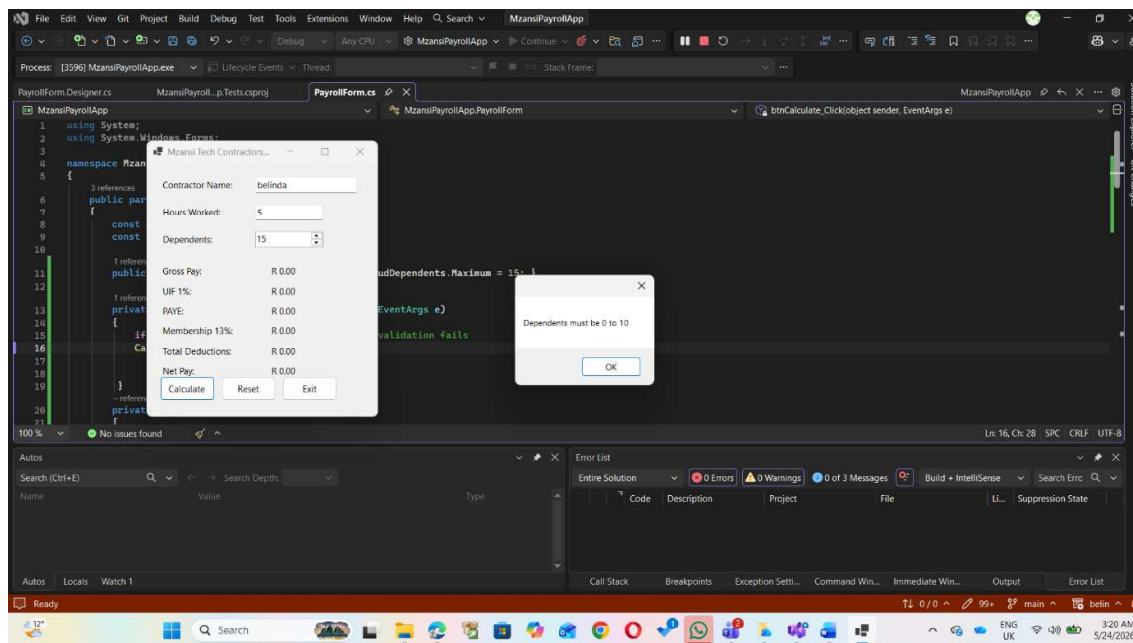


T3-non-numeric number





T4- No dependents > 10



The screenshot displays a Windows development environment with a Visual Studio IDE. In the foreground, a dialog box titled "MzansiTech Contractors..." is open, showing the following details:

- Contractor Name: Belinda
- Hours Worked: 40
- Dependents: 5
- Gross Pay: R 38000.00
- UIF 1%: R 380.00
- PAYE: R 6768.75
- Membership 13%: R 4940.00
- Total Deductions: R 12088.75
- Net Pay: R 25911.25

The background shows the Visual Studio IDE with the "PayrollForm.cs" file open. The code defines a "CalculatePay()" method that calculates the net pay based on the input values. The code includes comments for each deduction and a final "totalDed" calculation.

```

43: public void CalculatePay()
44: {
45:     gross = hours * HOURLY_RATE;
46:     uif = gross * 0.01; // 1%
47:     membership = gross * 0.13; // 13%
48:
49:     // PAYE = (Gross - (Gross * 0.0575 + Dependents)) * 25%
50:     payeBase = gross - (gross * 0.0575m + dependents);
51:     if (payeBase < 0) payeBase = 0;
52:     paye = payeBase * 0.25m;
53:
54:     totalDed = uif + paye + membership;
55: }

```

The "Autos" pane shows the following variable values:

Name	Value	Type
gross	38000.00	double
uif	380.00	double
membership	4940.00	double
payeBase	6768.75	double
paye	1692.19	double
totalDed	12088.75	double

The "Error List" pane is empty, and the "Autos" pane shows the following variable values:

Name	Value	Type
gross	38000.00	double
uif	380.00	double
membership	4940.00	double
payeBase	6768.75	double
paye	1692.19	double
totalDed	12088.75	double

Conclusion

System Testing found 1 defect in Dependents validation where values >10 were auto-corrected. After applying the fix and temporary adjustment of Maximum, Retesting confirmed the validation popup now appears for 15. Regression Testing with valid data T6 confirmed no impact on payroll calculation. All PM04 requirements are satisfied.